Question

Iodometry is a very commonly used analytical method in analytical chemistry. Based on the type of titration, it can be divided into direct iodometry and indirect iodometry. Among the following ten substances, some are suitable for direct iodometry, while others are suitable for indirect iodometry.

1. Cl_2
2. N_2H_4
3. O_3
4. NO_2^-
5. HCN
6. H_3AsO_3
7. H_3AsO_4
8. SO_2
9. Ce^{4+}
10. C_2H_5SH

Regarding the type of iodometry suitable for these ten substances and the pH conditions during the titration process, the following statements are correct:

A. Substances suitable for direct iodometry have even serial numbers.

B. Substances suitable for indirect iodometry have two sequence numbers that are not odd.

C. There are six types of substances suitable for indirect iodometry.

D. Substance No. 2 needs to be titrated in an acidic environment.

E. Substance No. 9 can be titrated under weakly alkaline conditions.

F. Substance No. 4 can be titrated under weakly alkaline conditions.

G. None of the above statements are correct

Answer

Correct Answer: G

Detailed Explanation

The difference between direct iodometry and indirect iodometry is that direct iodometry uses a standardized I_2 standard solution to directly titrate the substance, using excess iodine to indicate the endpoint; while indirect iodometry uses excess I^- to react with oxidizing substances to generate I_2, and then uses sodium thiosulfate solution to titrate I_2. Thus, the various substances are analyzed as follows:

CHECKPOINT 1 PTS

Direct iodometry uses a standardized I_2 standard solution to directly titrate

CHECKPOINT 1 PTS

Indirect iodometry uses excess I^- to react with oxidizing substances to generate I_2, and then uses sodium thiosulfate solution to titrate I_2

1. Cl_2: It cannot react directly with elemental iodine, but it has strong oxidizing properties and can oxidize I^-, thus it uses indirect iodometry.

CHECKPOINT 1 PTS

Cl_2 has strong oxidizing properties and can oxidize I^-, thus it uses indirect iodometry

2. N_2H_4: It has strong reducing properties and can be oxidized by I_2, thus it uses direct iodometry. Since the titration of N_2H_4 will produce hydrogen ions:

$$N_2H_4 + 2I_2 = N_2 + 4H^+ + 4I^-$$

The reaction produces hydrogen ions, so it should not be titrated in an acidic environment, but should be titrated under weakly alkaline buffer conditions, so option D is incorrect.

CHECKPOINT 1 PTS

N_2H_4 has strong reducing properties and can be reduced by I_2, thus it uses direct iodometry

CHECKPOINT 1 PTS

$N_2H_4 + 2I_2 = N_2 + 4H^+ + 4I^-$ produces hydrogen ions, and should be titrated under weakly alkaline buffer conditions

3. O_3 has strong oxidizing properties and can oxidize I^-, thus it uses indirect iodometry.

CHECKPOINT 1 PTS

O_3 has strong oxidizing properties, thus it uses indirect iodometry

4. NO_2^- has oxidizing properties and can oxidize I^-, thus it uses indirect iodometry. This reaction cannot be carried out under alkaline conditions, because the oxidation products of NO_2^- under neutral/weakly alkaline conditions are not fixed, and may be ammonia or nitrogen gas; but under acidic conditions, the oxidation product is fixed as nitric oxide, so option F is incorrect.

CHECKPOINT 1 PTS

NO_2^- has oxidizing properties, thus it uses indirect iodometry

CHECKPOINT 1 PTS

The oxidation products of NO_2^- under neutral/weakly alkaline conditions are not fixed, and titration should be performed under acidic conditions

5. HCN can be oxidized by elemental iodine to produce the pseudohalogen ICN, so it can be titrated by direct iodometry.

CHECKPOINT 1 PTS

HCN can be oxidized by elemental iodine to produce the pseudohalogen ICN, thus it uses direct iodometry

6/7. H_3AsO_3 has reducing properties, and arsenic in the +3 valence state will be oxidized to +5 by elemental iodine. Similarly, arsenic in the +5 valence state, H_3AsO_4, can oxidize I^- and be reduced to +3. Therefore, H_3AsO_3 uses direct iodometry, and H_3AsO_4 uses indirect iodometry.

CHECKPOINT 1 PTS

H_3AsO_3 in the +3 valence state will be oxidized to +5 by elemental iodine. Similarly, arsenic in the +5 valence state, H_3AsO_4, can oxidize I^- and be reduced to +3

8. SO_2 can be oxidized to sulfate by I_2, so direct iodometry is used.

CHECKPOINT 1 PTS

SO_2 can be oxidized to sulfate by I_2, thus it uses direct iodometry

9. Ce^{4+} is a strong oxidant, and uses indirect iodometry to oxidize I^-. Ce^{4+} will turn into a precipitate under alkaline conditions, so it cannot be titrated under weakly alkaline conditions, so option E is incorrect.

CHECKPOINT 1 PTS

Ce^{4+} is a strong oxidant, thus it uses indirect iodometry

CHECKPOINT 1 PTS

Ce^{4+} will turn into a precipitate under alkaline conditions, and cannot be titrated under weakly alkaline conditions

10. Thiols have reducing properties and can form disulfide bonds, so I_2 oxidation is used, which is direct iodometry.

CHECKPOINT 1 PTS

Thiols have reducing properties and can form disulfide bonds, thus I_2 oxidation is used, which is direct iodometry

In summary, direct iodometry includes 2, 5, 6, 8, 10; indirect iodometry includes 1, 3, 4, 7, 9, so options A, B, and C are all incorrect.

CHECKPOINT 1 PTS

Direct iodometry includes 2, 5, 6, 8, 10; indirect iodometry includes 1, 3, 4, 7, 9

In summary, options A-F are all incorrect, and option G is correct.